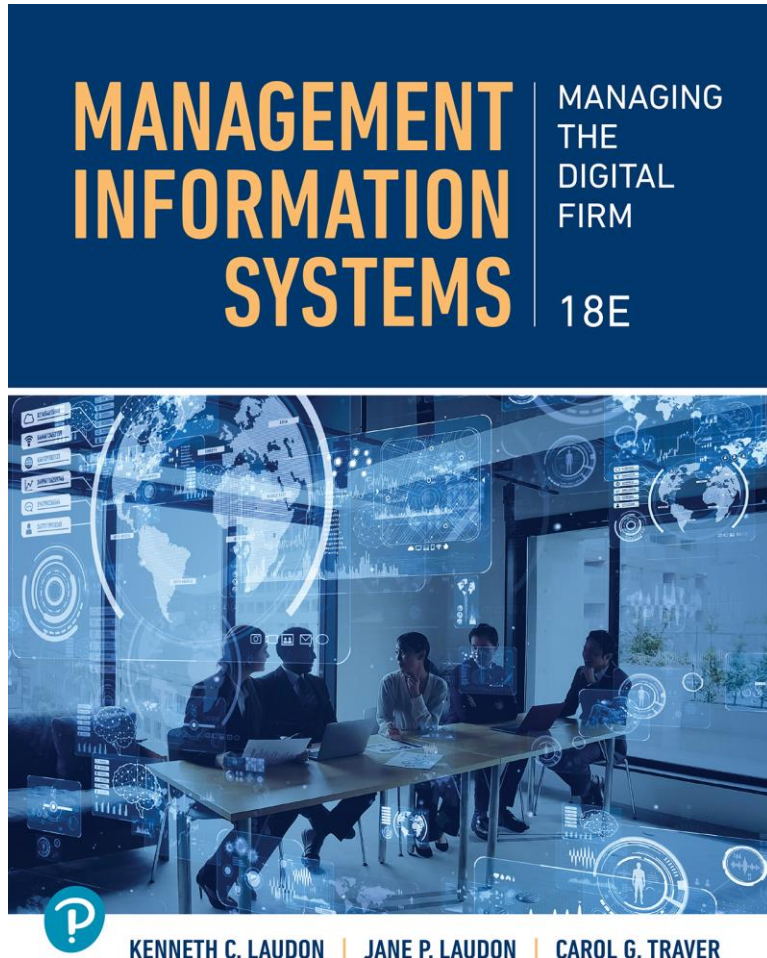


Management Information Systems: Managing the Digital Firm

Eighteenth Edition



Chapter 12

Enhancing Decision Making with Data Analytics and Business Intelligence

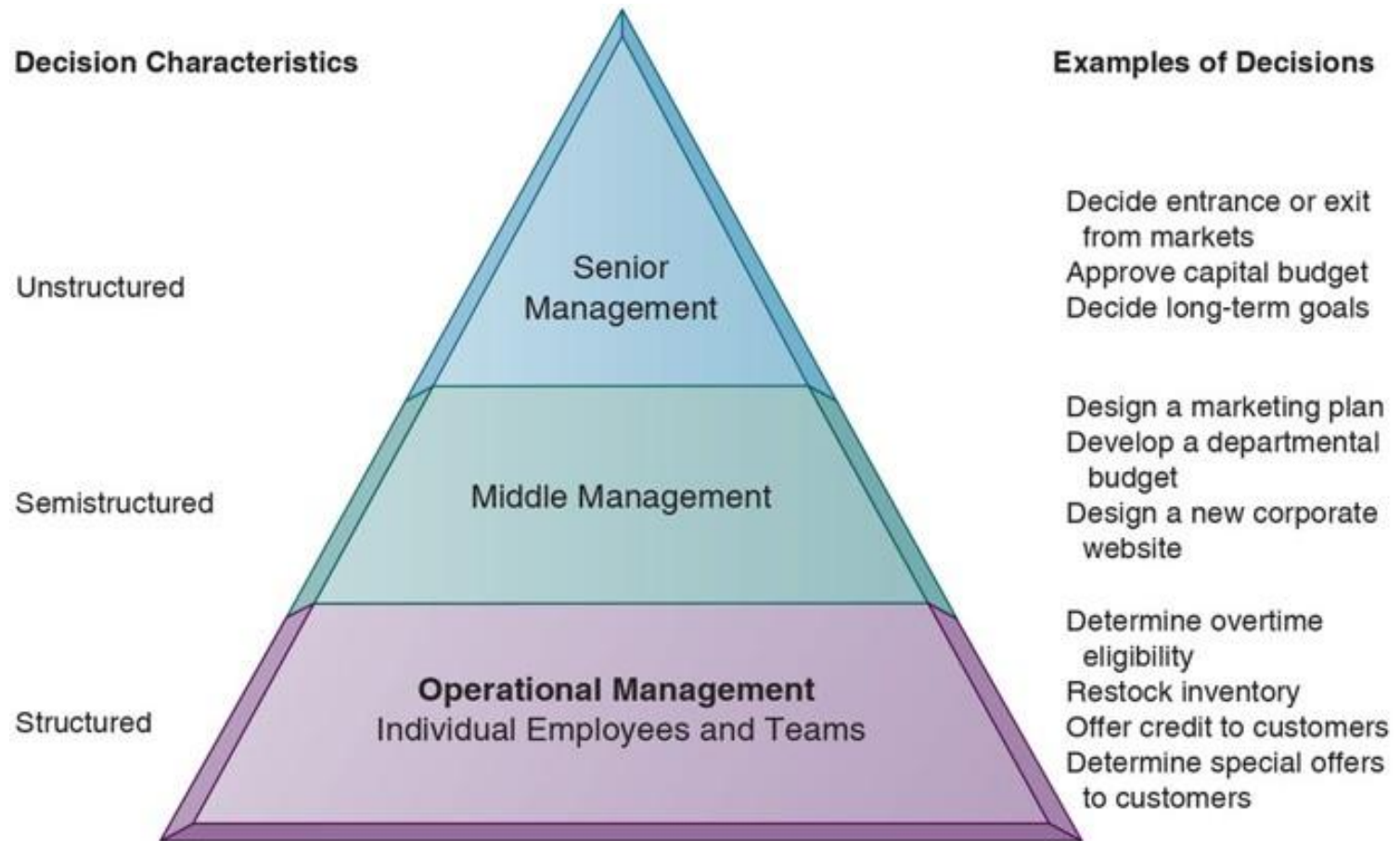
Learning Objectives

- 12.1** Describe the different types of business decisions.
- 12.2** Describe the decision-making process.
- 12.3** Describe the different roles managers play.
- 12.4** Understand the value of improved decision making.
- 12.5** Understand why IS do not always result in better decisions.
- 12.6** Discuss how data analytics, BI and AI support decision making.
- 12.7** Describe the capabilities of BI systems.
- 12.8** Understand the different types of BI users.
- 12.9** Understand how the information in this chapter can help your career.

Different Types of Business Decisions

- Types of business decisions
 - **Unstructured**
 - Decision maker must provide judgment, evaluation, and insight to solve problem
 - **Structured**
 - Repetitive and routine; involve definite procedure for handling so they do not have to be treated each time as new
 - **Semistructured**
 - Only part of problem has clear-cut answer provided by accepted procedure

Figure 12.1 Information Requirements of Key Decision-Making Groups in a Firm

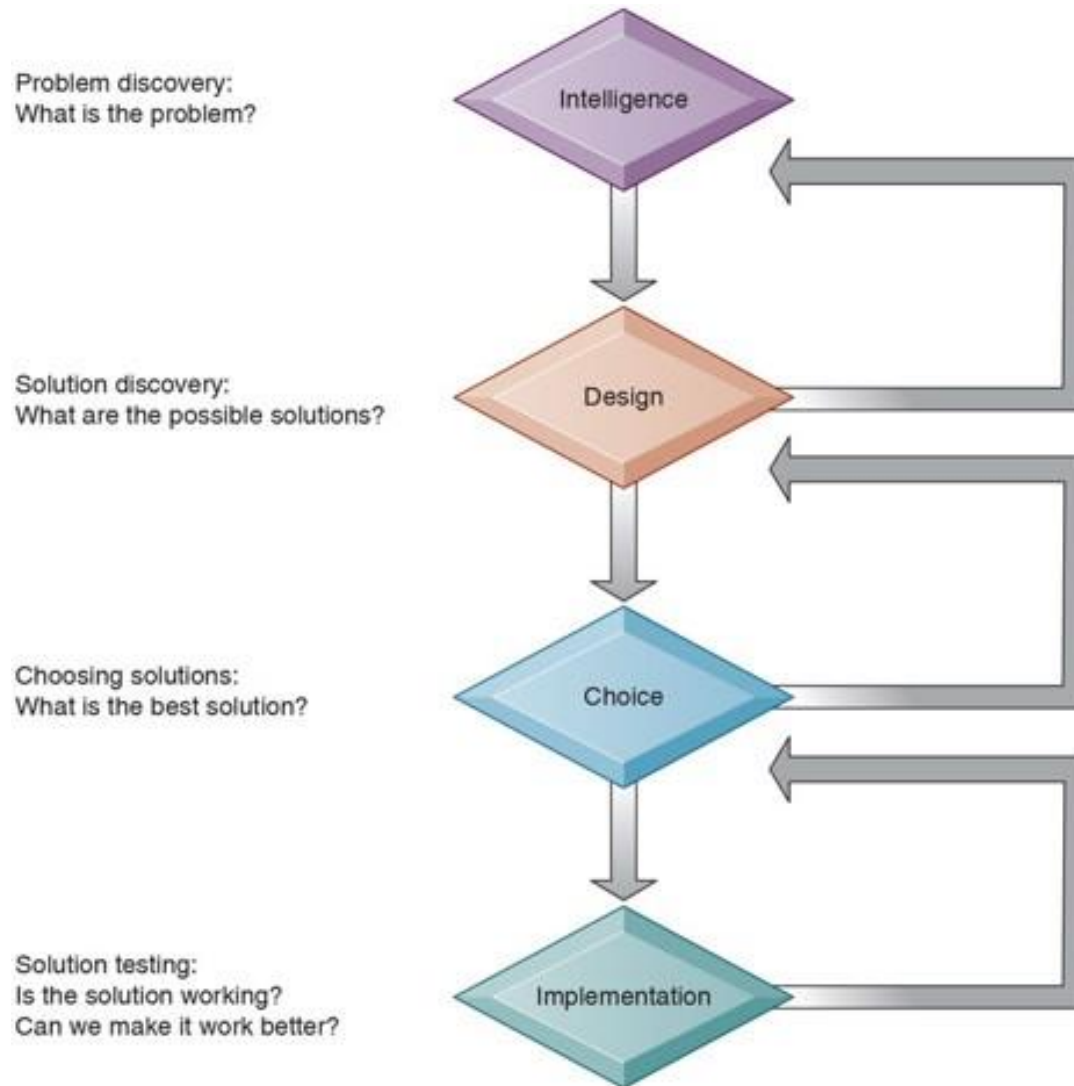


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Stages of Decision-Making Process

- Making a decision is a multistep process
- Simon (1960) described four different stages in decision making: intelligence, design, choice, and implementation
 - **Intelligence** - Discovering, identifying, and understanding the problems occurring in the organization
 - **Design** - Identifying and exploring solutions to the problem
 - **Choice** - Choosing among solution alternatives
 - **Implementation** - Making chosen alternative work and continuing to monitor how well solution is working

Figure 12.2 Stages in Decision Making



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Different Roles Managers Play (1 of 3)

- Systems to support decision making produce
 - Better decision making by managers and employees
 - Above-average returns on investment (ROI) for the firm
 - Ultimately, higher profitability

Different Roles Managers Play (2 of 3)

- **Classical model of management: five functions**
 - Planning, organizing, coordinating, deciding, and controlling
 - **Behavioral model of management**
 - Argues that the actual behavior of managers appears to be:
 - less systematic,
 - more informal,
 - less reflective,
 - more reactive, and
 - less well organized
- than the classical model would have us believe

Different Roles Managers Play (3 of 3)

- Managerial roles
 - Expectations of the activities that managers should perform in an organization
- Mintzberg found that these managerial roles fell into three categories
 - Interpersonal
 - Informational
 - Decisional

Table 12.1**Managerial Roles and Supporting Information Systems**

Role	Behavior	Support Systems
Interpersonal Roles	Interpersonal	
Figurehead		Telepresence systems
Leader		Telepresence, social networks
Liaison		Smartphones, social networks
Informational Roles	Information processing	
Nerve center		Management information systems, executive support systems
Disseminator		Texting, email, social networks
Spokesperson		Webinars, telepresence
Decisional Roles	Decision making	
Entrepreneur		None exist
Disturbance handler		None exist
Resource allocator		Business intelligence, decision-support system
Negotiator		None exist

Authors and Mintzberg, Henry. "Managerial Work: Analysis from Observation." *Management Science* 18 (October 1971).



Value of Improved Decision Making

- One of the main contributions of information systems has been to improve decision making for both individuals and groups
 - What does it mean to a business to make better decisions?
 - What is the monetary value of improved decision making?

Table 12.2**Business Value of Improved Decision Making**

Example Decision	Decision Maker	Number of Annual Decisions	Estimated Value to Firm of a Single Improved Decision	Annual Value
Allocate support to most valuable customers	Accounts manager	12	\$100,000	\$1,200,000
Predict call center daily demand	Call center management	4	\$150,000	\$600,000
Decide parts inventory levels daily	Inventory manager	365	\$5,000	\$1,825,000
Identify competitive bids from major suppliers	Senior management	1	\$2,000,000	\$2,000,000
Schedule production to fill orders	Manufacturing manager	150	\$10,000	\$1,500,000
Allocate labor to complete a job	Production floor manager	100	\$4,000	\$400,000

Why Information Systems Do Not Always Result in Better Decisions

- Three main reasons why investments in IT do not always produce positive results
 - **Information quality**
 - High-quality decisions require high-quality information
 - **Management filters**
 - Managers have selective attention and have variety of biases that reject information that does not conform to prior conceptions
 - **Organizational inertia and politics**
 - Strong forces within organizations resist making decisions calling for major change

Table 12.3**Information Quality Dimensions**

Quality Dimension	Description
Accuracy	Do the data represent reality?
Integrity	Are the structure of data and relationships among the entities and attributes consistent?
Consistency	Are data elements consistently defined?
Completeness	Are all the necessary data present?
Validity	Do data values fall within defined ranges?
Timeliness	Are data available when needed?
Accessibility	Are the data accessible, comprehensible, and usable?

High-Velocity Automated Decision Making

- Made possible through computer algorithms precisely defining steps for a highly structured decision
 - Humans taken out of decision
- For example: High-speed computer trading programs
 - Trades executed in nanoseconds
- Require safeguards to ensure proper operation and regulation

Risks Introduced by Using AI in Decision Making

- Using AI for management decision making presents a number of challenges
- Risks include the potential for
 - Ingrained bias
 - Concerns about the data upon which the decisions are based
 - Accuracy
 - Lack of transparency as to how decisions are arrived at
 - Potential ethical issues

How Data Analytics, BI, and AI Support Decision Making (1 of 3)

- **Data analytics**

- Involves collecting, storing, inspecting, cleaning, modeling, and querying data

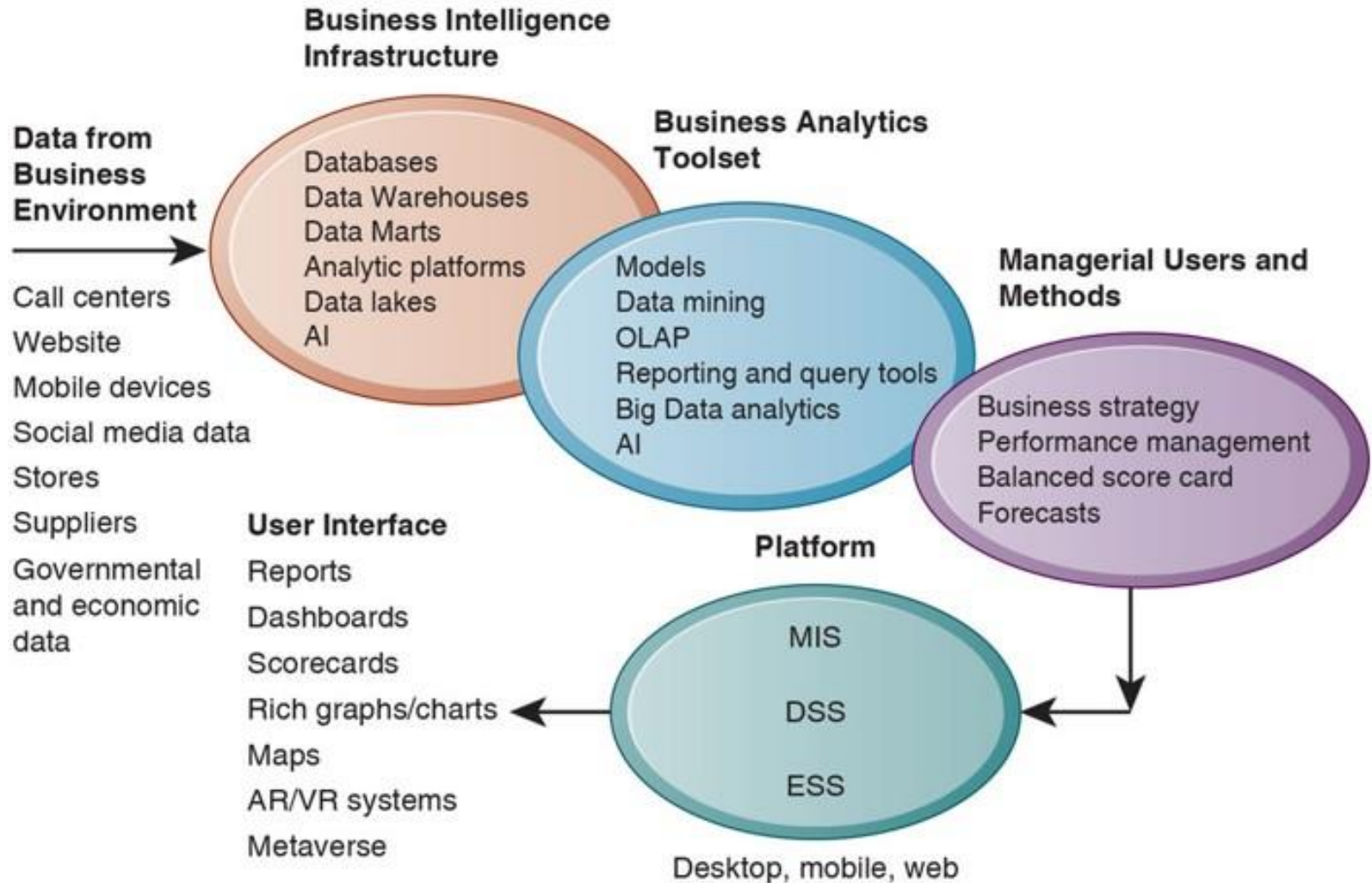
- **Business intelligence (BI)**

- Infrastructure for collecting, storing, and analyzing data produced by business
- Databases, data warehouses, data marts, Hadoop, analytic platforms

- **Business analytics (BA)**

- Tools and techniques for analyzing data
- OLAP, statistics, models, data mining

Figure 12.3 BI and Analytics for Decision Support



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How Data Analytics, BI, and AI Support Decision Making (2 of 3)

- Six elements in the business intelligence environment
 - Data from the business environment
 - BI infrastructure
 - BI and BA toolsets
 - Managerial users and methods
 - Delivery platform—MIS, DSS, ESS
 - User interface
 - Data visualization
 - Helps users see patterns and relationships in large amounts of data by presenting the data in graphical form

How Data Analytics, BI, and AI Support Decision Making (3 of 3)

- **BI software**

- Tableau Software is a leading BI and data visualization tool

- **Virtual reality (VR)**

- Powerful capabilities for three-dimensional data visualization
- VR systems use interactive graphics software and hardware
 - Creates computer-generated simulations that close to reality that users almost believe they are participating in a real-world situation

- **Augmented reality (AR)**

- A related technology for enhancing visualization by overlaying digital data and images onto a physical real-world environment

Capabilities of BI Systems

- Goal is to deliver accurate real-time information to decision makers
- Main analytic functionalities of BI systems
 - Production reports
 - Parameterized reports
 - Dashboards/scorecards
 - Ad hoc query/search/report creation
 - Drill down
 - Forecasts, scenarios, models

Table 12.4

Examples of BI Predefined Production Reports

Business Functional Area	Production Reports
Sales	Sales forecasts; sales team performance; cross-selling; sales cycle times
Service/call center	Customer satisfaction; service cost; resolution rates; churn rates
Marketing	Campaign effectiveness; loyalty and attrition; market basket analysis
Procurement and support	Direct and indirect spending; off-contract purchases; supplier performance
Supply chain	Backlog; fulfillment status; order cycle time; bill of materials analysis
Financial	General ledger; accounts receivable and payable; cash flow; profitability
Human resources	Employee productivity; compensation; workforce demographics; retention

Predictive and Prescriptive Analytics

- **Predictive analytics**

- Uses statistical analysis, data mining techniques, historical data, and assumptions about future conditions to **predict future trends and behavior patterns**

- **Prescriptive analytics**

- Differs from predictive analytics in not only predicting future outcomes but also **recommending actions or decisions to achieve desired outcomes or prevent undesirable ones**

Big Data Analytics

- Big data: Massive datasets collected from social media, online and in-store customer data, and so on
- Help create real-time, personalized shopping experiences for major online retailers
- Smart cities in the public sector
 - Public record data
 - Sensors, location data from smartphones
 - Ability to evaluate effect of one service change on system

Table 12.5

What Big Data Analytics Can Do

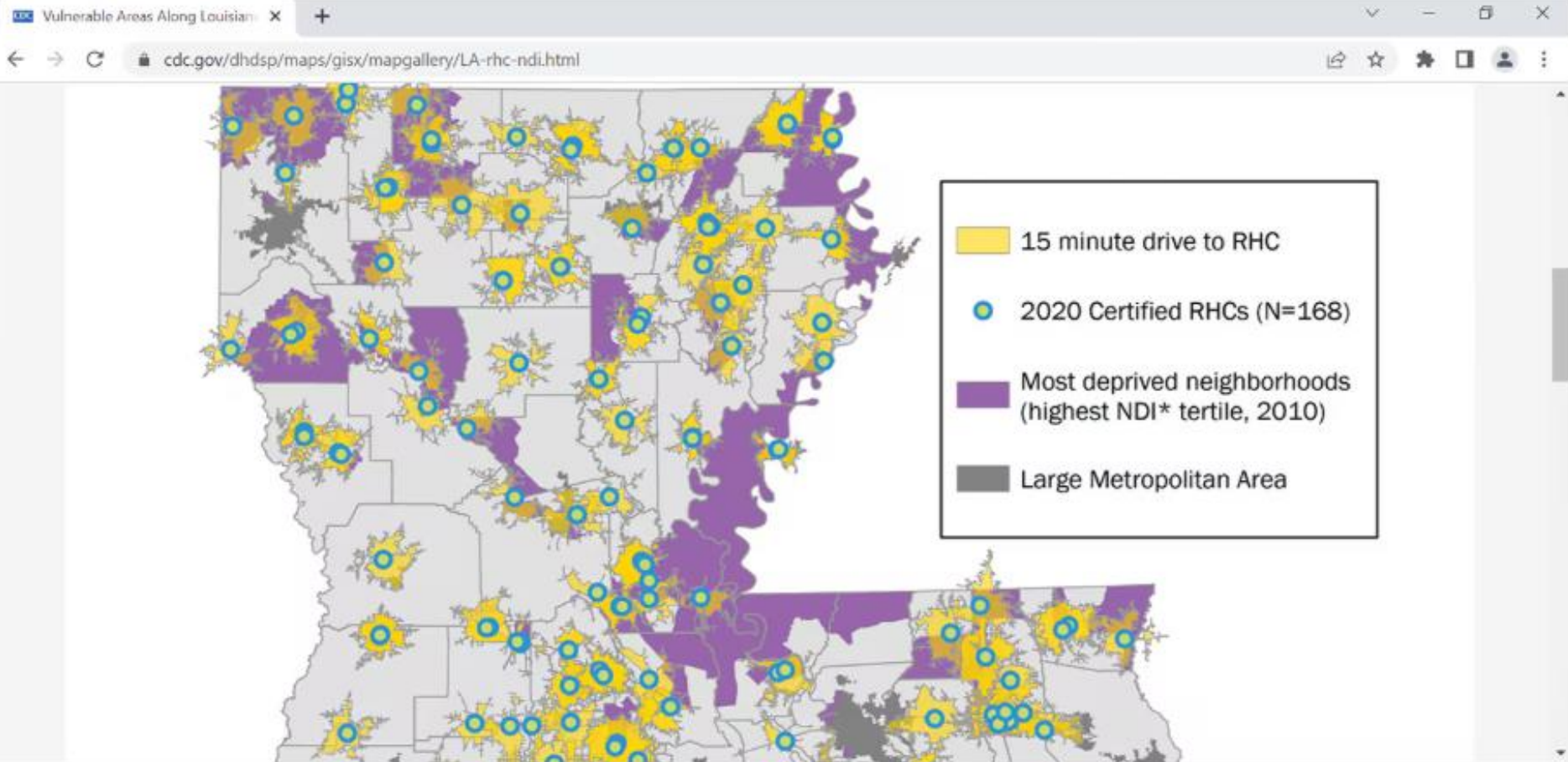
EHarmony	Online dating website analyzes personal and behavioral data on 10 million users to match couples based on features of compatibility found in thousands of successful relationships. Processes more than 15 million matches per day.
Rolls Royce	Uses big data to reduce the amount of carbon its aircraft engines produce while optimizing maintenance to prolong the life of an engine. Monitors how each engine flies, the conditions under which it is flying, and how pilots use it. Customizes maintenance regime for individual engines.
Hertz	Uses data from web surveys, email, text messages, website traffic patterns, and data gathered at all of its 8,500 rental car locations to adjust staffing levels at individual locations in order to accommodate times of the day when demand peaks or falls.
Levi Strauss & Co,	Levi's uses SAS Analytics to analyze millions of consumer demand signals to create a supply chain geared to the preferences of individual consumers. The system enables Levi's to create demand forecasts for each style, color, and size of product that it offers.

Operational Intelligence and Analytics

- Operational intelligence
 - A form of BI and BA that can be used to deliver insights about activities inside and outside the organization from these streams of big data as they are generated in real time
- Collection and use of data generated by sensors
- Internet of Things (IoT)
 - Creating huge streams of data from web activities, sensors, and other monitoring devices
- Software for operational intelligence and analytics enable companies to analyze their big data

Location Analytics and Geographic Information Systems

- Location analytics
 - Ability to gain business insight from the location (geographic) component of data
 - Mobile devices
 - Output from sensors, scanning devices
 - Map data
- Geographic information systems (GIS)
 - Ties location-related data to maps
 - Example: For helping local governments calculate response times to disasters



This map shows the locations of vulnerable areas along Louisiana's Mississippi Delta that lack geographic access to Rural Health Clinics (RHCs). RHCs provide primary care access points for rural area residents. Although many rural areas have access to nearby RHCs, there are still rural neighborhoods along the Mississippi Delta with no nearby clinics.

Source: Anna L. Hamilton, Well-Ahead Louisiana. Accessed from the Centers for Disease Control and Prevention's Chronic Disease GIS Exchange <https://www.cdc.gov/dhdsp/maps/gisx/mapgallery/LA-rhc-ndi.html>

Different Types of BI Users

- **Senior executives** use BI to monitor firm activities using visual interfaces like dashboards and scorecards
- **Middle managers and analysts** are more likely to be immersed in the data and software, entering queries and slicing and dicing the data along different dimensions
- **Operational employees** will, along with customers and suppliers, be looking mostly at prepackaged reports

Table 12.6**Examples of MIS Applications**

Company	MIS Application
Baylis & Harding	Wholesale distributor specializing in luxury toiletries and gift sets adopted BI tools from Phocas to enable better visibility into financial, customer and sales data. Managers and executives use the tools to create standard and ad hoc MIS reports based on data from the company's NetSuite ERP system. The reports provide instant access to the business data managers need to act proactively.
Black & Veatch	Global engineering, procurement, consulting, and consulting company uses MIS to track construction costs for various projects across the United States.
Chipotle Mexican Grill	Restaurant chain with more than 3,200 restaurants worldwide Implemented a BI system to track operational effectiveness. By standardizing reporting, the company was able to develop uniform KPIs for benchmarking.

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Figure 12.4 BI Users

**Power Users:
Producers
(20% of employees)**

Capabilities

**Casual Users:
Consumers
(80% of employees)**

IT developers

Production Reports

Customers/suppliers
Operational employees

Super users

Parameterized Reports

Senior managers

Business analysts

Dashboards/Scorecards

Managers/Staff

Analytical modelers

Ad hoc queries; Drill down
Search/OLAP

Business analysis

Forecasts; What if
Analysis; statistical models

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Support for Semistructured Decisions

(1 of 2)

- Decision-support systems
 - Support for semistructured decisions
 - Use mathematical or analytical models

Support for Semistructured Decisions

(2 of 2)

- Decision-support systems allow varied types of analysis
 - “What-if” analysis
 - Sensitivity analysis
 - A form of modeling that involves asking what-if questions repeatedly to predict a range of outcomes when one or more variables are changed multiple times
 - Backward sensitivity analysis
 - Multidimensional analysis / OLAP
 - For example: pivot tables

Figure 12.5 Sensitivity Analysis

Figure 12.5

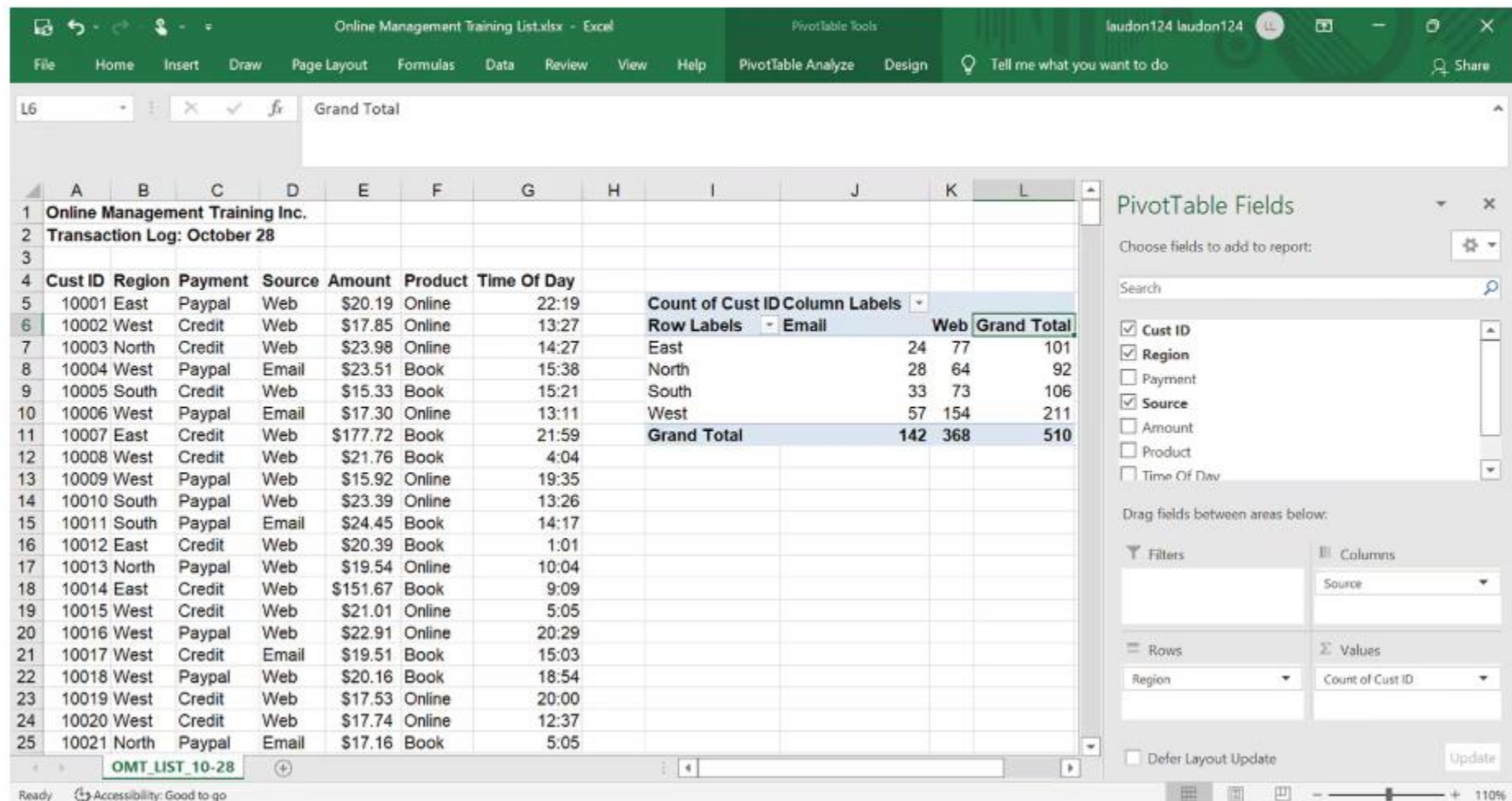
Sensitivity Analysis

Total fixed costs	19000					
Variable cost per unit	3					
Average sales price	17					
Contribution margin	14					
Break-even point	1357					
		Variable Cost per Unit				
Sales	1357	2	3	4	5	6
Price	14	1583	1727	1900	2111	2375
	15	1462	1583	1727	1900	2111
	16	1357	1462	1583	1727	1900
	17	1267	1357	1462	1583	1727
	18	1188	1267	1357	1462	1583

This table displays the results of a sensitivity analysis of the effect of changing the sales price of a beach towel and the cost per unit on the product's break-even point. It answers the question, "What happens to the break-even point if the sales price and the cost to make each unit increase or decrease?"

Figure 12.6

A Pivot Table That Examines Customer Regional Distribution and Advertising Source



This pivot table enables you to examine where an online management training company's customers come from in terms of region and advertising source.

Decision Support for Senior Management: Balanced Scorecard and Business Performance Management Methods (1 of 3)

- **ESS:** decision support for senior management
 - Help executives focus on important performance information
- **Balanced scorecard method**
 - Measures outcomes on four dimensions
 - Financial
 - Business process
 - Customer
 - Learning and growth
 - Key performance indicators (KPIs) measure each dimension

Figure 12.7 The Balanced Scorecard Framework



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Decision Support for Senior Management: Balanced Scorecard and Business Performance Management Methods (2 of 3)

- **Key performance indicators (KPIs)**
 - Measures proposed by senior management for understanding how well the firm is performing along any given dimension
- **Business performance management (BPM)**
 - Translates firm's strategies (e.g., differentiation, low-cost producer, scope of operation) into operational targets

Decision Support for Senior Management: Balanced Scorecard and Business Performance Management Methods (3 of 3)

- Data for a firm's ESS are supplied by the firm's existing enterprise applications:
 - Enterprise resource planning
 - Supply chain management
 - Customer relationship management

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